

Group Number: 1

LIBRARY BOOK TRACKING SYSTEM

Project Proposal

Problem Description

Libraries manage large collections of books and serve numerous members, making manual tracking of book issuances, returns, and member records a challenging and error-prone task. Without a proper system, it becomes difficult to keep track of which books are available, which have been issued, to whom they have been given, and when they are due for return.

This often leads to mismanagement, loss of books, delayed returns, and difficulties in calculating fines for overdue items. The Library Book Tracking System is designed to solve these problems by providing a structured, automated solution that enables librarians to efficiently add and search books, issue and return them, monitor due dates, manage member records, and automatically calculate fines ensuring smooth and organized library operations.

Proposed Features

- Add and search books
- Issue and return books
- Track due dates
- Manage member records
- Basic fine calculation

Group Members

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Source Code

The following C++ source code implements the Library Book Tracking System using Object-Oriented Programming concepts including Inheritance and Constructors.

```
#include <iostream>
#include <string>
using namespace std;

// ===== BASE CLASS
class Book {
protected:
    string title;
public:
    Book() {
        title = "";
    }
    Book(string t) {
        title = t;
    }
    string getTitle() {
        return title;
    }
    void display() {
        cout << title;
    }
};

// ===== DERIVED CLASS
class IssuedBook : public Book {
public:
    string studentID;

    // Default constructor
    IssuedBook() {
        title = "";
        studentID = "";
    }

    // Parameterized constructor
    IssuedBook(string t, string id) {
        title = t;
        studentID = id;
    }
};

// ===== LIBRARY CLASS
class Library {
private:
    static const int MAX = 50;
```

```

Book availableBooks[MAX];
int availCount;
IssuedBook issuedBooks[MAX];
int issuedCount;

public:
    Library() {
        availCount = 0;
        issuedCount = 0;
    }

    void addBook(string t) {
        if (availCount < MAX) {
            availableBooks[availCount] = Book(t);
            availCount++;
        }
    }

    void searchAndIssue() {
        string title;
        cout << "Enter book title: ";
        cin.ignore();
        getline(cin, title);
        int foundIndex = -1;

        for (int i = 0; i < availCount; i++) {
            if (availableBooks[i].getTitle() == title) {
                foundIndex = i;
                break;
            }
        }

        if (foundIndex == -1) {
            cout << "Book not available.\n";
            return;
        }

        cout << "Book is available.\n";
        cout << "Press 1 to issue: ";
        int choice;
        cin >> choice;
        if (choice != 1) {
            cout << "Issue cancelled.\n";
            return;
        }

        string id;
        cout << "Enter Student ID: ";
        cin >> id;

        issuedBooks[issuedCount] = IssuedBook(title, id);
        issuedCount++;
    }

```

```

        for (int i = foundIndex; i < availCount - 1; i++) {
            availableBooks[i] = availableBooks[i + 1];
        }
        availCount--;

        cout << "Book issued successfully.\n";
        cout << "Return within 15 days.\n";
    }

void returnBook() {
    string id;
    cout << "Enter Student ID: ";
    cin >> id;
    cout << "\nBooks issued to " << id << ":\n";
    bool found = false;
    for (int i = 0; i < issuedCount; i++) {
        if (issuedBooks[i].studentID == id) {
            cout << "- " << issuedBooks[i].getTitle() << endl;
            found = true;
        }
    }

    if (!found) {
        cout << "No books found for this ID.\n";
        return;
    }

    string title;
    cout << "Enter book title to return: ";
    cin.ignore();
    getline(cin, title);
    int returnIndex = -1;

    for (int i = 0; i < issuedCount; i++) {
        if (issuedBooks[i].studentID == id && issuedBooks[i].getTitle() ==
title) {
            returnIndex = i;
            break;
        }
    }

    if (returnIndex == -1) {
        cout << "Book not found.\n";
        return;
    }

    int days;
    cout << "After how many days are you returning the book? ";
    cin >> days;

    int fine = 0;
    if (days > 15) {
        fine = (days - 15) * 50;
    }
}

```

```

    }
    cout << "Fine: Rs. " << fine << endl;

    addBook(issuedBooks[returnIndex].getTitle());

    for (int i = returnIndex; i < issuedCount - 1; i++) {
        issuedBooks[i] = issuedBooks[i + 1];
    }
    issuedCount--;
    cout << "Book returned successfully.\n";
}

void displayBooks() {
    if (availCount == 0) {
        cout << "No books available.\n";
        return;
    }
    cout << "\nAvailable Books: \n";
    for (int i = 0; i < availCount; i++) {
        cout << i + 1 << ". ";
        availableBooks[i].display();
        cout << endl;
    }
}

};

int main() {
    Library lib;
    lib.addBook("Quran");
    lib.addBook("Bible");
    lib.addBook("Torah");
    lib.addBook("C++ Primer");
    lib.addBook("Data Structures");

    int choice;
    do {
        cout << "\n===== LIBRARY MENU =====\n";
        cout << "1. Search & Issue Book\n";
        cout << "2. Return Book\n";
        cout << "3. Add Book\n";
        cout << "4. Display Available Books\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1: lib.searchAndIssue(); break;
            case 2: lib.returnBook(); break;
            case 3: {
                string t;
                cout << "Enter title: ";
                cin.ignore(); getline(cin, t);
                lib.addBook(t);
            }
        }
    } while (choice != 5);
}

```

```
        break;
    }
    case 4: lib.displayBooks(); break;
    case 5: cout << "Goodbye!\n"; break;
}
} while (choice != 5);
return 0;
}
```

Final Output / Results of the Program

The following console outputs demonstrate the system's functionality during runtime.

Scenario 1: Search & Issue a Book

```
===== LIBRARY MENU =====
1. Search & Issue Book
2. Return Book
3. Add Book...
Enter choice: 1
Enter book title: C++ Primer
Book is available.
Press 1 to issue: 1
Enter Student ID: F25M-BSAI-040
Book issued successfully.
Return within 15 days.
```

Scenario 2: Return a Book (On Time)

```
Enter choice: 2
Enter Student ID: F25M-BSAI-040
Books issued to F25M-BSAI-040:
- C++ Primer
Enter book title to return: C++ Primer
After how many days are you returning the book? 10
Fine: Rs. 0
Book returned successfully.
```

Scenario 3: Return a Book (Overdue Fine Example)

```
Enter Student ID: F25-BSAI-037
Books issued to F25-BSAI-037:
- Algorithms
Enter book title to return: Algorithms
After how many days are you returning the book? 18
Fine: Rs. 150
Book returned successfully.
```

Scenario 4: Display Available Books

```
Enter choice: 4
Available Books:
1. Quran
2. Bible
3. Torah
4. Data Structures
5. Artificial Intelligence
```

Scenario 5: Exit Program

```
Enter choice: 5
Good Bye!
```